

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 8: DQB Synthesis & Assessment — Driving Question Resolution and Unit Wrap-Up

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	D	D	D	C	D

USE AFTER: OBSERVE PHASE — Observe: Benedict's test practical on the simulated 'digested ugali' sample (Steps 1–4)

Q1. A learner is doing Benedict's test on a sample. Which set of steps is in the correct order?

- A. Put about 2 cm³ of test solution in a test tube, add an equal amount of Benedict's solution and mix, place the tube in a hot water bath for a few minutes, then observe and record colour changes**
- B. Add Benedict's solution to a dry test tube, add a few drops of test solution, mix, and record the colour immediately without heating
- C. Put about 2 cm³ of test solution in a test tube, place it in a hot water bath, then add Benedict's solution and record the colour once the tube is cold
- D. Heat the Benedict's solution alone, add about 2 cm³ of test solution, then record the colour change without mixing

Correct answer: A. The four steps are: put about 2 cm³ of test solution in a tube, add an equal amount of Benedict's solution and mix, heat in a hot water bath for a few minutes, then observe and record all colour changes. Heating is required for the reaction, and mixing comes before heating.

USE AFTER: OBSERVE PHASE — Observe: Step 4 – observing and recording the colour change in the Benedict's test

Q2. After heating a glucose solution with Benedict's solution in a hot water bath, a brick-red precipitate forms. What does this result show?

- A. The solution contains a reducing sugar such as glucose**
- B. The solution has no sugar, because the blue colour has been lost
- C. The solution contains protein, because Benedict's solution reacts with all nutrients
- D. The solution contains starch, because starch turns Benedict's solution red

Correct answer: A. A brick-red (orange-red) precipitate is a positive result for reducing sugars such as glucose. Starch and proteins are not detected by Benedict's test, and a blue colour would mean no reducing sugar.

USE AFTER: OBSERVE PHASE — Observe: written question (a) – why the test solution represents digested ugali rather than whole ugali starch

Q3. In the practical, a dilute glucose solution was used to represent 'digested ugali'. Why is glucose a suitable choice to represent digested ugali?

- A. Ugali starch is already a reducing sugar, so it gives a positive Benedict's test before any digestion
- B. Ugali contains mainly protein, and glucose is the end-product of protein digestion
- C. Hydrochloric acid in the stomach turns starch into glucose within seconds
- D. Amylase enzymes break the large starch molecules in ugali down into simple sugars such as glucose**

Correct answer: D. Ugali is mostly starch. Amylase breaks starch down into maltose and then glucose, so digested ugali contains glucose, which Benedict's test can detect. Whole starch does not give a positive result.

USE AFTER: EXPLAIN PHASE — Explain: pairs' written answer to the driving question – naming enzyme-region matches with correct substrates and products

Q4. Nyama choma is rich in protein. Which statement correctly matches the enzyme pepsin to its region of action and substrate?

- A. Pepsin acts in the mouth and breaks down starch into maltose
- B. Pepsin acts in the small intestine and breaks down fats into fatty acids and glycerol
- C. Pepsin acts in the stomach and breaks down starch into glucose
- D. Pepsin acts in the stomach and breaks down proteins into smaller peptides**

Correct answer: D. Pepsin is secreted in the stomach, works in the acidic conditions there, and breaks proteins such as those in nyama choma into smaller peptides. Starch is digested by amylase and fats by lipase.

USE AFTER: EXPLAIN PHASE — Explain: sentence on what happens when an adaptation fails (e.g. ulcers caused by H. pylori disrupting the stomach's mucus lining)

Q5. In the Explain phase, pairs wrote about what happens when an adaptation of the digestive system fails. Infection with the bacterium H. pylori can damage the mucus lining of the stomach. Why can this lead to an ulcer?

- A. The stomach starts to digest starch, which burns a hole in the wall
- B. The villi in the stomach are destroyed, so nutrients cannot be absorbed
- C. The stomach can no longer make any acid, so food stays undigested and forms a sore
- D. The wall of the stomach loses its protection and is damaged by the acid and pepsin in the stomach juice**

Correct answer: D. The mucus lining protects the stomach wall from hydrochloric acid and pepsin. When H. pylori damages the mucus, the wall is exposed and gets damaged, forming an ulcer. This also links to the puzzle of how the body avoids digesting itself.

USE AFTER: MODEL BUILDING PHASE — Model Building: comparing the Lesson 1 journey map with the Lesson 7 3D model – linking to model label 'villi'

Q6. Sukuma wiki, ugali and nyama choma are broken down into small soluble molecules. How do the villi and microvilli of the ileum help the body absorb these nutrients efficiently?

- A. They make the intestine wall thick so that nutrients cannot escape back into the food
- B. They grind the food into smaller pieces so it can pass through the intestine wall
- C. They increase the surface area of the intestine wall, so more nutrients can be absorbed into the rich network of capillaries**
- D. They release hydrochloric acid to dissolve the nutrients so they can enter the blood

Correct answer: C. Villi and microvilli greatly increase the absorbing surface area of the ileum. Each villus has a rich capillary network that carries absorbed nutrients away in the blood.

USE AFTER: END OF LESSON — End of lesson: after the Model Building final sentence 'The human digestive system is adapted ... because...' and the journey map upgrades

Q7. A learner's Lesson 1 sentence said, 'Enzymes break down food.' Which upgraded sentence is the most scientifically complete?

- A. Enzymes in the mouth and stomach help to break down food into smaller parts.
- B. Pepsin in the mouth breaks down the proteins in ugali.
- C. The mouth breaks down ugali when we chew it with our teeth.
- D. Salivary amylase in the mouth begins to break down the starch in ugali into maltose.**

Correct answer: D. A complete sentence names the enzyme, where it acts, the substrate and the product: salivary amylase, mouth, starch in ugali, maltose. The other options are vague or wrongly matched (chewing is mechanical digestion, and pepsin works in the stomach).